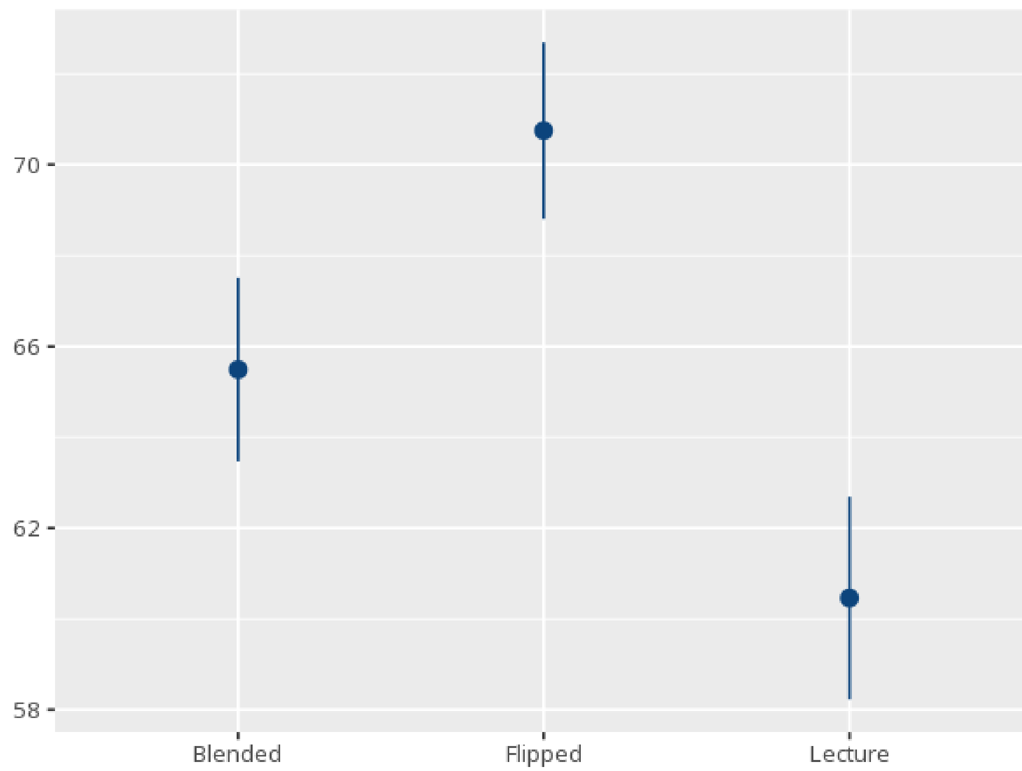
1 One-way ANOVA + post-hoc

Group	N	Mean	Std. Dev.
Blended	80	65.49	6.85
Flipped	80	70.75	6.58
Lecture	80	60.46	7.57

Source	SS	df	MS	F	p	η^2 [99% CI]
Between	4234.40	2	2117.20	43.07	<.001	0.27 [0.16, 1.00]
Within	11650.16	237	49.16			

Pair	Mdiff	SE	padj	99% CI
Blended - Flipped	−5.26	1.11	<.001	[−8.52, −2.00]
Blended - Lecture	5.03	1.11	<.001	[1.77, 8.29]
Flipped - Lecture	10.29	1.11	<.001	[7.03, 13.55]



The F and its p tell you whether the groups differ overall; η^2 (or ω^2) is the effect size. If significant, the post-hoc table shows which specific pairs differ, with multiplicity-adjusted p-values. Your significance threshold (α) is 0.05.

A one-way ANOVA gave F(2,237)=43.07, $p < .001$, η^2=.27 [.16, 1.00]. Tukey HSD post-hoc tests showed. . .

Statistical basis: F-test for equality of several means (ANOVA) (Fisher, R. A. (1925). Statistical Methods for Research Workers. Oliver & Boyd.); Tukey HSD post-hoc comparison (Tukey, J. W. (1949). "Comparing individual means in the analysis of variance." Biometrics, 5(2), 99-114.); Bonferroni-corrected post-hoc comparison (a 'posthoc' choice) (Dunn, O. J. (1961). "Multiple comparisons among means." Journal of the American Statistical Association, 56(293), 52-64.); Scheffé-corrected post-hoc comparison (a 'posthoc' choice) (Scheffé, H. (1953). "A method for judging all contrasts in the analysis of variance." Biometrika, 40(1-2), 87-104.); Levene's test for equal variances (Brown-Forsythe median-centered variant) (Levene, H. (1960). "Robust tests for equality of variances." In I. Olkin (Ed.), Contributions to Probability and Statistics (pp. 278-292). Stanford University Press.); Brown-Forsythe median-centered variant of the homogeneity-of-variance test (Brown, M. B., Forsythe, A. B. (1974). "Robust tests for the equality of variances." Journal of the American Statistical Association, 69(346), 364-367.); η^2 effect-size benchmarks (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.).